

Supplementary Information for Decision Relevance of Uncertainty Calibration in Active-Learning Materials Screening

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1 Scope and reproducibility provenance

This Supplementary Information documents the confirmation analysis used in the accompanying manuscript. The numerical authority is the D-drive D-only confirmation package generated on 25 September 2026. The package contains 3,216 JSON records corresponding to the complete four-task, two-split, 40-seed design. The validation report records 3,216 expected and 3,216 actual JSON files, with no missing or invalid records. The analysis manifest reports 577,298,134 bytes across the JSON records.

The manuscript intentionally does not pool these results with earlier schedule or calibration analyses. Those historical outputs remain useful as project lineage, but their relation to the corrected acquisition semantics has not been re-established by a source-hash, configuration, and trajectory audit. They are therefore not evidence for the six primary contrasts in this manuscript. This separation prevents a numerical namespace generated under an earlier analysis path from being silently treated as part of the current confirmation.

The frozen protocol signatures are:

- analysis core SHA-256: 84e364e6a6671f9b5ef3a563cb7c93fc1a321f01f05c64ce52a8277f50f72111;
- experiment-plan SHA-256: 3ae049e83417a8b3ee350e7b48dab88c1998ec4b129904183c571228c2f0c3ec;
- confirmation-freeze SHA-256: c050f374796ed3d5a68819bb0d7b6b436947545ef48948b9ca29942d51760f88;
- selected calibration interpolation exponent: $\alpha = 0.5$.

2 Data partitions and campaign geometry

The four Matbench task suffixes were `expt_gap`, `mp_gap`, `log_gvrh`, and `log_kvrh`. This set was inherited from the frozen benchmark protocol; a separate pre-confirmation task-selection rationale is not recoverable from the archived materials. Each task used `iid_record` and `chemical_system_ood` splits. The latter is a true group-disjoint chemical-system split under the inherited protocol; it was not replaced by an IID fallback or a capped approximation. The target pool contained no initially labelled target candidates, and unqueried target labels were unavailable to fitting and acquisition. Queried target labels were added to the labelled training set after each batch; the complete target labels were used only for retrospective endpoint evaluation.

The surrogate was an 80-estimator extra-randomized tree ensemble with minimum leaf size two. The acquisition objective was maximization. Each campaign used the protocol-defined

initial labelled set, four query batches, and a 20% target-pool query budget. All conditions within a paired seed shared the same task, split, initial state, model fitting seeds, batch geometry, and stable identifier tie rule whenever the condition did not deliberately alter uncertainty assignment.

2.1 Feature representation and preprocessing

The surviving historical adapter documents Pymatgen composition featurization: parse formulae with `pymatgen.core.Composition`, construct a lexicographically sorted element vocabulary for the task, encode atomic fractions, and append total atom count and distinct-element count. Exact historical runtime identity remains unverified because the V5 source lock and feature cache are unavailable. In V2, Pymatgen 2024.10.29 (Ong et al., 2013) produced dense float64 matrices with 83 columns for `matbench_expt_gap` and 86 for each remaining task. Missing values were filled with zero; non-finite values were rejected. No scaling, target transformation, feature selection, constant-feature removal, or categorical encoding was applied. The vocabulary used formulas across the complete unlabeled task snapshot before splitting, without reading targets; this is an unsupervised transductive schema step. No persistent feature cache was used. The per-task column names and matrix SHA-256 values are in the V2 data manifest.

3 Calibration implementation

At each campaign state, the labelled training set comprised the initial source labels and labels revealed by preceding target queries. Valid calibration rows required at least three OOB predictions, and the calibrator required at least eight valid rows. OOB ensemble predictions supplied μ_j^{OOB} and σ_j^{OOB} for currently labelled candidates. The residual ratio was

$$q_j = \frac{|y_j - \mu_j^{\text{OOB}}|}{\max(\sigma_j^{\text{OOB}}, 10^{-8})}. \quad (1)$$

An ExtraTrees model with 64 estimators, minimum leaf size five, all available features, and a state-derived random seed fitted $\log(q_j + 10^{-8})$. Balanced five-fold OOF predictions were formed by a deterministic material-ID hash order. Writing $g_j = \log(q_j + 10^{-8})$, the local multiplier was

$$c = \frac{\text{median}\{q_j / \exp(\hat{g}_j^{\text{OOF}})\}}{0.67448975}. \quad (2)$$

The corrected local scale for a candidate was $s_{\text{local}}(x) = c \exp(\hat{g}(x))$, clipped to $[0.25, 4]$. The positive global scale was $s_{\text{global}} = \max(10^{-8}, \text{median}(q_j) / 0.67448975)$ over valid OOB rows; the implementation used the identity scale if fewer than eight valid rows were available. The local clipping precedes interpolation, so the final interpolated scale is not separately constrained to $[0.25, 4]$. The confirmation scale was the geometric interpolation

$$s_\alpha(x) = \exp[(1 - \alpha) \log s_{\text{global}} + \alpha \log s_{\text{local}}(x)], \quad \alpha = 0.5. \quad (3)$$

The development stage selected $\alpha = 0.5$ using fixed-state MACE and a prespecified tolerance rule. The confirmation endpoint was not used to select α , and no optional stopping was permitted.

The historical local implementation is retained in the raw project for provenance but is not used for the current LOCAL claim. In the current namespace, `LOCAL` refers only to the corrected

OOF implementation described above; `LEGACY_LOCAL` is retained as a diagnostic condition and is not one of the six primary contrasts.

4 Acquisition conditions

The rank and magnitude scores were

$$A_{\text{rank}}(x) = r(\mu(x)) + r(\sigma(x)), \quad (4)$$

$$A_{\text{mag}}(x) = \mu(x) + \sigma(x). \quad (5)$$

The rank function used average percentile ranks, and the selection order used stable material identifiers for deterministic tie resolution. RAW used the unmodified ensemble uncertainty. GLOBAL applied a positive constant to uncertainty and was included for magnitude acquisition as a control in the complete matrix, while the independent selected-ID identity check concerned rank acquisition.

PERMUTED independently permuted the local scale vector within each seed and campaign stage. It therefore preserved the local-scale distribution but removed its candidate-specific allocation. MATCHED was defined separately for the two acquisition families. For magnitude acquisition, its scalar uncertainty weight matched the root-mean-square local uncertainty bonus to the raw bonus. For rank acquisition, a fixed grid searched for the weight whose next-batch disagreement from RAW matched the LOCAL-versus-RAW disagreement. The tie rule was closest to one, then lower weight. The residual matching gap was recorded in each run file.

5 Primary aggregation

For each condition, the run-level endpoint was $L_{4,\text{query}}$. The analysis first formed paired differences within each task-split environment for each seed, then took the unweighted mean across the eight environments. The resulting 40 seed-level means were the only values used for primary inference. Run-file counts and environment-row counts are not substitute sample sizes.

The six primary contrasts and their complete results are shown in Table S1. The archived plan records a descriptive practical threshold of 0.005 but does not preserve its scientific rationale; it is not an externally validated minimal important difference. Percentile bootstrap confidence intervals used 10,000 draws with seed 926900. Sign-flip p-values used 20,000 draws with seed 926901. Holm adjustment was applied across the six two-sided sign-flip tests.

Table S1: Complete primary contrast summary.

Rule	Contrast	LOCAL mean	Comparator mean	Difference	95% CI	Holm p_{adj}	Positive
Rank	RAW	0.536415	0.517452	0.018964	[0.012242,0.025864]	0.000300	31/40
Rank	PERMUTED	0.536415	0.469933	0.066482	[0.059380,0.073640]	0.000300	40/40
Rank	MATCHED	0.536415	0.523306	0.013110	[0.005787,0.020719]	0.001800	27/40
Magnitude	RAW	0.618213	0.603772	0.014441	[0.008274,0.020594]	0.000300	29/40
Magnitude	PERMUTED	0.618213	0.602535	0.015679	[0.010217,0.020958]	0.000300	33/40
Magnitude	MATCHED	0.618213	0.603517	0.014696	[0.010357,0.019227]	0.000300	35/40

Each row is LOCAL minus the named comparator. Means are paired-seed means across eight fixed environments. Positive counts refer to seed-level differences greater than zero.

5.1 Descriptive task-split heterogeneity

The pooled seed-level effect concealed substantial heterogeneity across the fixed task-split environments. Table S2 reports the mean LOCAL-RAW difference within each environment and the number of positive seed-level differences. These rows are descriptive decompositions of the same 40 paired seeds used in the primary analysis; they are not additional independent replicates and no multiplicity claim is attached to them. The rank effect was positive in both splits for the experimental gap, logarithmic shear modulus, and logarithmic bulk modulus tasks, whereas it was negative for both splits of the materials-project gap task. Magnitude acquisition showed the same broad pattern, with a smaller negative mean for the IID materials-project gap split and a larger negative mean for its OOD split.

Table S2: Descriptive LOCAL-RAW heterogeneity across the eight fixed task-split environments.

Task	Split	Rank acquisition		Magnitude acquisition	
		Mean difference	Positive/40	Mean difference	Positive/40
expt_gap	IID	0.01565	27/40	0.01256	27/40
expt_gap	Chemical OOD	0.00824	22/40	0.00819	26/40
log_gvrh	IID	0.03044	29/40	0.02146	27/40
log_gvrh	Chemical OOD	0.03280	25/40	0.02493	25/40
log_kvrh	IID	0.07365	34/40	0.02464	26/40
log_kvrh	Chemical OOD	0.06382	29/40	0.03498	29/40
mp_gap	IID	-0.03823	3/40	-0.00195	17/40
mp_gap	Chemical OOD	-0.03465	9/40	-0.00929	16/40

Mean difference is LOCAL minus RAW in $L_{4,\text{query}}$. Positive/40 counts seed-level differences greater than zero. The eight rows are fixed environments, not independent population samples.

6 Condition-level descriptive summaries

For reference, the 320 environment rows per condition yielded the following means. These values are descriptive and should not be analysed as 320 independent replicates. L1 is the mean diagnostic MACE over the four campaign states; it differs from the shared RAW-state calibration summary below because condition trajectories can diverge.

Table S3: Descriptive condition means over 320 task-split-seed environment rows.

Rule	Condition	$L_{4,\text{query}}$	L1	Final recall	Runtime (s)
Rank	RAW	0.5175	0.0610	0.7751	32.50
Rank	LOCAL	0.5364	0.0401	0.8021	26.11
Rank	PERMUTED	0.4699	0.0468	0.7210	28.41
Rank	MATCHED	0.5233	0.0621	0.8022	27.67
Magnitude	RAW	0.6038	0.0645	0.9006	6.13
Magnitude	GLOBAL	0.6007	0.0473	0.9013	6.20
Magnitude	LOCAL	0.6182	0.0413	0.9032	25.25
Magnitude	PERMUTED	0.6025	0.0492	0.9049	26.14
Magnitude	MATCHED	0.6035	0.0644	0.9001	24.82

The large runtime dispersion is expected from heterogeneous benchmark sizes and is not a primary endpoint. In particular, condition-level means are not used to claim that calibration reduces computational cost.

7 Calibration diagnostic summary

The fixed-state calibration file contains 1,280 records per diagnostic method. Mean MACE was 0.0610 for RAW, 0.0465 for GLOBAL, 0.0524 for OOF- $\alpha = 1$, and 0.0412 for OOF- $\alpha = 0.5$. The corresponding medians were 0.0526, 0.0403, 0.0446, and 0.0358. These confirmation-state values describe the frozen intervention; the interpolation exponent had already been selected using separate development seeds. They are not retrospectively used to choose the exponent. They are not a new independent confirmatory test.

Supplementary Figure S3 and Table S5 report the mean observed interval coverage at each nominal level. Each point averages 1,280 shared fixed-state records. These descriptive means do not show the distribution across states. MACE was calculated as an absolute deviation within each state before averaging, so the mean-coverage curve need not reproduce the method ordering by mean MACE.

The Gaussian NLL means in the diagnostic CSV are extremely heavy-tailed, with means near 1.8×10^{12} and medians near one. Because this discrepancy indicates sensitivity to a small number of extreme records, the manuscript does not use mean Gaussian NLL to rank methods or support a calibration claim. The raw diagnostic CSV is retained in the project package for transparent audit.

8 Additional descriptive decision summaries

The initial-state batch disagreement was $1 - |S_{\text{RAW}} \cap S_{\text{LOCAL}}|/k$, where each set contained the top k candidates under its acquisition score on the same initial model and pool. The manuscript reports means and medians over eight environments and 40 seeds for each rule. All 320 expected initial-state records were finite and included; no rows were excluded. This summary was added during manuscript revision and is descriptive, with no new multiplicity-adjusted hypothesis test. Later-stage disagreement values were not pooled into this comparison because campaign states can diverge after the first query batch.

8.1 Exploratory association between initial MACE and campaign utility

For each task-split-seed block, we calculated the stage-0 change in mean absolute coverage error (MACE) as OOF-LOCAL ($\alpha = 0.5$) minus RAW from the shared-state diagnostic, and the campaign utility change as LOCAL minus RAW in $L_{4,\text{query}}$. The two acquisition rules share the same stage-0 MACE value; their campaign utility differences are rule-specific. Table S4 reports the mean changes and Spearman correlation across 40 paired seeds within each fixed environment. The correlations are descriptive, with no p-values or population-level task interpretation. Figure 4 in the main text visualizes the four MP-gap conditions. These summaries show whether initial-state calibration changes co-occurred with later utility changes, but cannot determine which candidate errors or displacements generated the outcomes.

Exact invariance check. Across 16 task-split-seed cases, rank-GLOBAL selected the same material IDs as RAW. This deterministic check concerns implementation only.

Table S4: Exploratory association between stage-0 MACE change and campaign utility change across paired seed blocks.

Task	Split	Policy	Mean Δ MACE	Mean $\Delta L_{4,\text{query}}$	Spearman ρ
Experimental gap	IID	Rank	0.00844	0.01565	-0.12
Experimental gap	IID	Magnitude	0.00844	0.01256	-0.13
Experimental gap	Chemical OOD	Rank	0.00640	0.00824	0.11
Experimental gap	Chemical OOD	Magnitude	0.00640	0.00819	0.13
Materials Project gap	IID	Rank	-0.00631	-0.03823	0.03
Materials Project gap	IID	Magnitude	-0.00631	-0.00195	0.14
Materials Project gap	Chemical OOD	Rank	-0.00879	-0.03465	0.08
Materials Project gap	Chemical OOD	Magnitude	-0.00879	-0.00929	0.02
Log shear modulus (GVRH)	IID	Rank	-0.08087	0.03044	0.09
Log shear modulus (GVRH)	IID	Magnitude	-0.08087	0.02146	-0.23
Log shear modulus (GVRH)	Chemical OOD	Rank	-0.05978	0.03280	-0.03
Log shear modulus (GVRH)	Chemical OOD	Magnitude	-0.05978	0.02493	-0.02
Log bulk modulus (KVRH)	IID	Rank	-0.06787	0.07365	-0.04
Log bulk modulus (KVRH)	IID	Magnitude	-0.06787	0.02464	0.10
Log bulk modulus (KVRH)	Chemical OOD	Rank	-0.05427	0.06382	0.16
Log bulk modulus (KVRH)	Chemical OOD	Magnitude	-0.05427	0.03498	0.05

Each row contains 40 paired confirmation-seed blocks within one fixed task-split environment. Negative Δ MACE indicates lower MACE; negative $\Delta L_{4,\text{query}}$ indicates lower LOCAL utility. The correlations are exploratory summaries only; no significance tests are reported. Source: `tables/exploratory_mace_utility_association.csv`.

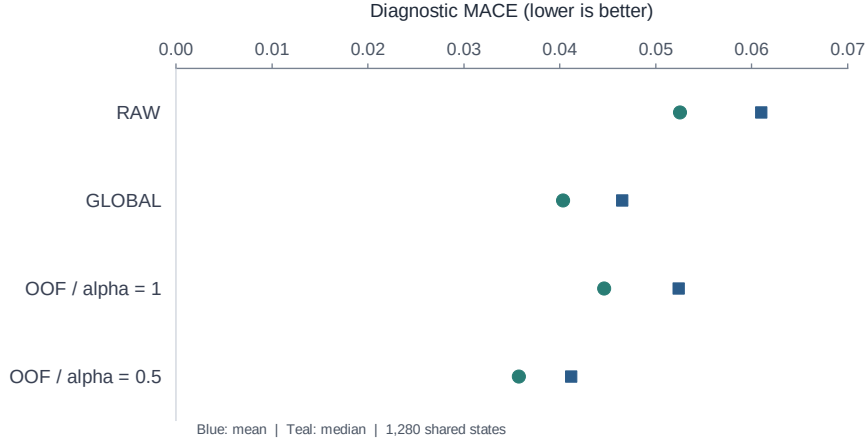


Figure S1: Fixed-state diagnostic MACE for RAW, GLOBAL, and corrected OOF local calibration at two frozen interpolation exponents. Blue squares denote means and teal circles medians over 1,280 shared-state records per method; lower values indicate smaller coverage error. This panel selects the four principal conditions discussed in the manuscript; all six diagnostic methods remain in the source table. The records are correlated campaign states and are not independent inferential replicates. No standard-error bars or new hypothesis tests are implied, and these confirmation diagnostics did not determine the exponent.

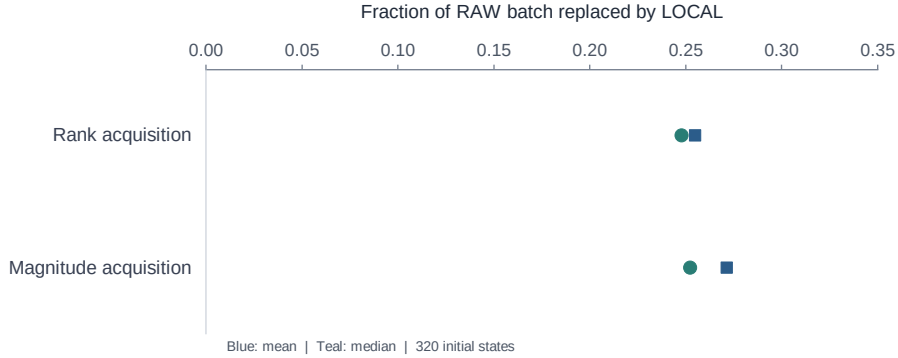


Figure S2: Initial-state selection changes after local calibration. Blue squares and teal circles show mean and median batch disagreement, respectively, over all 320 initial task-split-seed states for each acquisition rule. Disagreement is the fraction of the RAW batch replaced under LOCAL on the same initial model and pool. This post-analysis descriptive summary shows decision perturbation, not its benefit: discovery utility is assessed separately by the paired-seed campaign contrasts. No rows were excluded and no independent-replicate or significance claim is made.

The preceding coverage and MACE summaries assess marginal interval calibration on shared fixed states. They do not measure how well uncertainty orders prediction errors or whether queries improve the target-tail objective. As a separate exploratory diagnostic, we calculated candidate-level Spearman association between uncertainty and absolute prediction error on the retrospectively labelled V2 target candidate pools. In the LOCAL campaign states, the equal-weight mean association across eight fixed environments was 0.463 for rank and 0.454 for magnitude under corrected uncertainty, compared with 0.463 and 0.455 under RAW uncertainty (Table S6). Environment means ranged from 0.22 to 0.75. These candidate-pool values are not calibration estimates: candidate labels were attached retrospectively, the queried pool changes

Table S5: Mean observed interval coverage at each nominal level across shared fixed states.

Method	50% nominal	80% nominal	90% nominal	95% nominal
RAW	49.49%	76.22%	84.91%	89.56%
GLOBAL	53.59%	80.42%	87.93%	91.72%
LOCAL ($\alpha = 1$)	50.64%	77.14%	84.67%	88.75%
LOCAL ($\alpha = 0.5$)	52.39%	79.81%	87.35%	91.23%

Each value is the mean observed coverage among 1,280 correlated shared-state records. It is descriptive; no confidence interval or independent-replicate interpretation is implied.

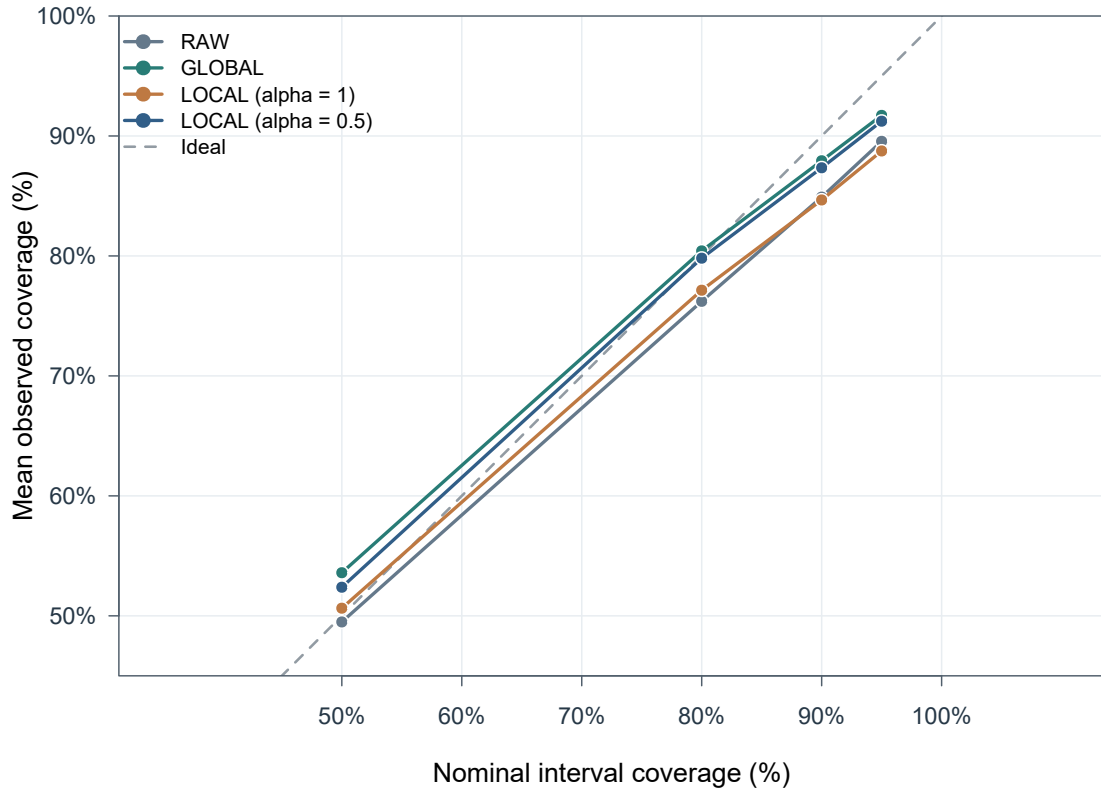


Figure S3: Mean observed interval coverage plotted against nominal coverage for the four principal fixed-state diagnostic methods. Each point averages 1,280 correlated campaign states. The dashed line denotes ideal mean coverage. These aggregated points do not display state-level variation, and the curve is not a substitute for the per-state absolute-deviation calculation used for MACE.

over stages, and the environments are fixed. Median and 90th-percentile absolute standardized residuals are summarized descriptively without a Gaussian distribution claim. Marginal calibration, uncertainty-error ranking, and discovery utility are distinct outcomes.

Table S6: Exploratory uncertainty–error alignment on V2 LOCAL trajectory candidate pools.

Policy	Scale	Mean ρ_s	Environment range	Median $ e /s$	Mean $q_{0.90}(e /s)$
Magnitude	LOCAL	0.454	[0.223, 0.750]	0.567	1.770
Magnitude	RAW	0.455	[0.202, 0.758]	0.717	2.120
Rank	LOCAL	0.463	[0.242, 0.747]	0.562	1.779
Rank	RAW	0.463	[0.220, 0.757]	0.702	2.095

Each row averages 640 seed–stage summaries with equal weight across eight fixed task–split environments (20 seeds and four stages per environment). The environment range is the minimum and maximum of the eight environment means, not an interval. Spearman ρ_s compares uncertainty with absolute prediction error. Standardized residual summaries use $|y - \mu|/s$ and make no Gaussian claim. Target labels were attached retrospectively and were not used in fitting or selection. These values are exploratory and are not independent inferential replicates.

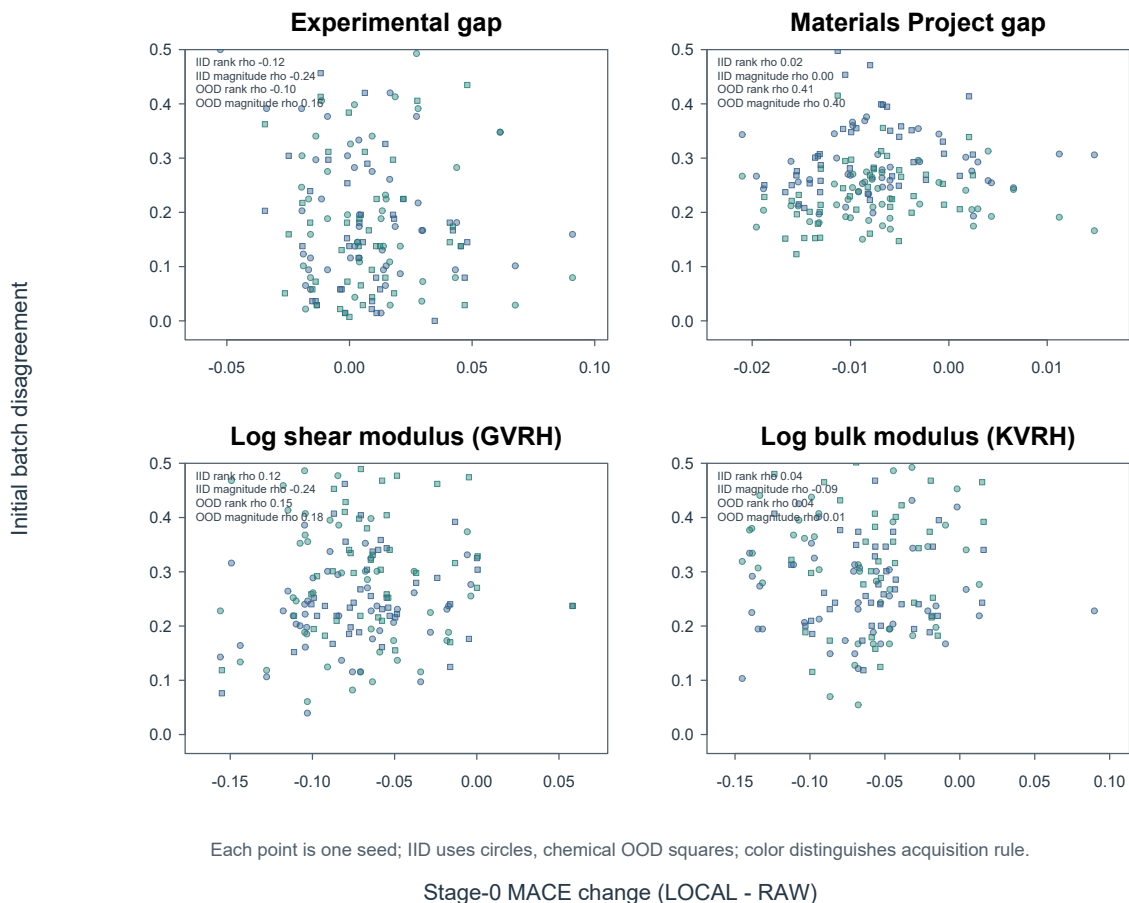


Figure S4: Exploratory association between stage-0 MACE change (OOF-LOCAL, $\alpha = 0.5$, minus RAW) and initial-batch disagreement (the fraction of RAW first-batch candidates absent from the LOCAL first batch). Each point is one paired seed within a task-split-policy environment ($n = 40$); circles denote IID, squares chemical OOD, blue denotes rank acquisition, and teal magnitude acquisition. The annotations are descriptive Spearman correlations. No p-values, independent-task inference, or candidate-level mechanism is implied.

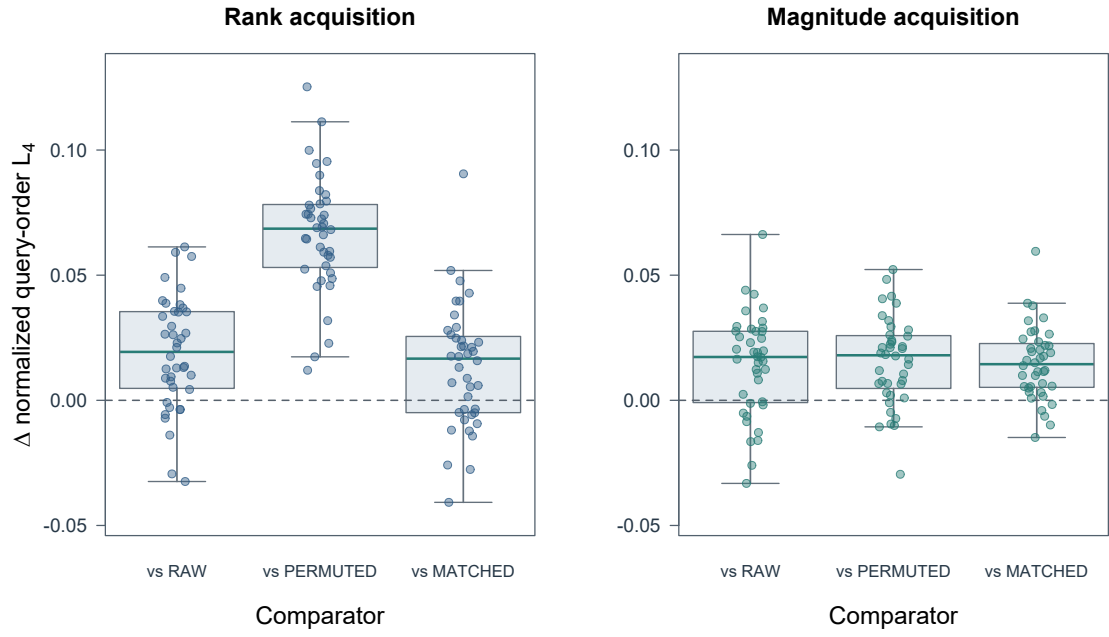


Figure S5: Distribution of the 40 paired-seed differences for each of the six prespecified primary contrasts in integrated top-1% recall over the query budget ($L_{4,\text{query}}$). Each point is one seed-level difference, calculated after averaging equally over the eight fixed task-split environments. Boxes show the median and interquartile range; whiskers extend to the most extreme observations within 1.5 interquartile ranges. The horizontal dashed line marks zero, and positive differences favor LOCAL. These points are the paired blocks used in the primary inference, not additional independent observations.

9 Exploratory candidate-value displacement analysis

The confirmation files retain ordered selected material IDs at each query stage. We used these IDs to compare the candidates entering the LOCAL batch with the candidates displaced from the RAW batch at the same stage, then joined their observed target values from the archived task tables. For each task–split–policy, we averaged the four stage-level entering-minus-displaced differences within each of the 40 paired seeds. Table S7 reports the mean target values and the distribution of these seed-block summaries. The join is retrospective: observed labels were not used to select candidates.

For `matbench_mp_gap`, rank-policy LOCAL entrants had higher mean target values than same-stage RAW-displaced candidates in both splits (+0.0673 IID and +0.1249 chemical OOD), whereas the magnitude-policy differences were slightly negative (−0.0162 and −0.0143). Since primary $L_{4,\text{query}}$ was lower under LOCAL for both rules in both splits, mean target values of exchanged batch members do not explain the top-1% utility loss. For these historical confirmation files, the target-pool membership and full candidate-level prediction vectors were not retained, so their entrant-tail enrichment cannot be reconstructed. The separate verified V2 files retain the relevant candidate-state fields and are analysed below. No causal or new confirmatory claim is made.

Table S7: Exploratory same-stage candidate displacement: observed target values of LOCAL-only entrants and RAW-only candidates.

Task	Split	Policy	LOCAL / RAW mean y	Mean Δy (SD)	Seed blocks $\Delta y < 0$
Experimental gap	Chemical OOD	Magnitude	1.796 / 1.883	-0.087 (0.307)	57%
Experimental gap	Chemical OOD	Rank	1.866 / 1.743	0.124 (0.326)	35%
Experimental gap	IID	Magnitude	1.856 / 1.852	0.003 (0.336)	48%
Experimental gap	IID	Rank	1.901 / 1.756	0.145 (0.296)	30%
Log shear modulus (GVRH)	Chemical OOD	Magnitude	1.783 / 1.862	-0.078 (0.048)	92%
Log shear modulus (GVRH)	Chemical OOD	Rank	1.790 / 1.781	0.010 (0.039)	38%
Log shear modulus (GVRH)	IID	Magnitude	1.779 / 1.853	-0.073 (0.045)	100%
Log shear modulus (GVRH)	IID	Rank	1.772 / 1.778	-0.005 (0.034)	50%
Log bulk modulus (KVRH)	Chemical OOD	Magnitude	2.068 / 2.169	-0.101 (0.059)	95%
Log bulk modulus (KVRH)	Chemical OOD	Rank	2.078 / 2.080	-0.002 (0.039)	55%
Log bulk modulus (KVRH)	IID	Magnitude	2.063 / 2.171	-0.108 (0.063)	98%
Log bulk modulus (KVRH)	IID	Rank	2.084 / 2.077	0.006 (0.043)	45%
Materials Project gap	Chemical OOD	Magnitude	2.811 / 2.826	-0.014 (0.140)	52%
Materials Project gap	Chemical OOD	Rank	2.793 / 2.668	0.125 (0.147)	25%
Materials Project gap	IID	Magnitude	2.971 / 2.988	-0.016 (0.065)	65%
Materials Project gap	IID	Rank	2.928 / 2.861	0.067 (0.076)	15%

Rows summarize 40 seed blocks per task–split–policy. Within each seed, differences were averaged equally across stages. Δy is LOCAL-only entrants minus same-stage RAW-only candidates, in native target units; labels were joined retrospectively. Descriptive only, no tests. Frozen target-pool membership is absent, so top-1% enrichment cannot be measured. Source: `tables/exploratory_selection_displacement_summary.csv`.

10 SECONDARY ROBUSTNESS ANALYSIS (V2)

The original six contrasts remain the sole confirmatory analysis. V2 is an independent secondary analysis, not a replication of historical V5. It uses an independently hashed snapshot of the four Matbench material tables, separate deterministic split-membership records, seeds 2026092600–2026092619, and an isolated output namespace. Each task–split–seed campaign uses the same source/diagnostic/target partition proportions (20/20/60), source-only initial training labels, a

target-pool query budget of 20%, and four balanced sequential batches.

The second randomized-tree-ensemble construction uses `RandomForestRegressor` (Breiman, 2001) with 80 bootstrap trees, minimum leaf size two, all features considered per split, and one worker. Raw uncertainty is the population standard deviation of the tree predictions. Out-of-bag tree predictions on currently labelled rows provide residual ratios for calibration; at least three OOB predictions are required per row, and at least eight rows are required to fit. The local-scale ExtraTrees model uses 64 trees, minimum leaf size five, five deterministic folds ordered by SHA-256 material-ID hashes, median normalization, clipping to $[0.25, 4]$, and the fixed geometric interpolation exponent $\alpha = 0.5$. The resulting correction multiplies raw uncertainty.

Candidate-level state files include the candidate ID, predicted mean, raw and corrected uncertainty, local scale, both counterfactual policy scores/ranks/selections, actual query order, and target value attached retrospectively after the batch was fixed. Target labels are excluded from fitting, calibration, and acquisition; diagnostic labels are used only for descriptive calibration summaries. The campaign code also records MACE on the fixed diagnostic partition, defined as the unweighted mean absolute deviation between empirical and nominal central-interval coverage at 50%, 80%, 90%, and 95%. RAW and LOCAL MACE are stored at each of four stages; the seed-level averages are auxiliary descriptive diagnostics, separate from the V2 utility endpoint. The complete per-stage candidate-state records are retained as Parquet files. Alternative target-tail fractions (0.5%, 1%, and 5%) are evaluated by post-processing the same campaigns; no threshold-specific model reruns are performed. No V2 p-values are planned; effect distributions and task/split heterogeneity are reported descriptively.

Exploratory MP-gap diagnostics. Candidate-level prediction-error alignment, tail-scale allocation, and LOCAL-only versus RAW-only candidate displacement are labelled exploratory. These retrospective diagnostics do not support causal chemical-mechanism claims.

10.1 V2 results

The mean top-1% LOCAL-RAW differences, after equal weighting across task-split environments, were +0.0057 for rank acquisition and +0.0149 for magnitude acquisition. Tail sensitivity was not monotonic across policies: rank showed a near-zero mean at 0.5% and a positive mean at 5%, whereas magnitude was positive at 0.5% and 1% and negative at 5%. Figure S8 and Table S10 show the task-split detail. All effects are descriptive paired summaries over 20 seeds; no V2 hypothesis tests were performed.

Initial RAW-LOCAL first-batch disagreement averaged 25.8% under rank and 34.6% under magnitude when task-split means were equally weighted (Table S11). The fixed diagnostic-partition MACE summaries are reported separately in Table S12; they are auxiliary calibration diagnostics and are not target-pool utility measures. Exploratory mean target values among queried candidates and MP-gap candidate-level diagnostics are in Tables S14 and S13, respectively. Queried-target averages do not measure target-tail enrichment, and candidate-level associations do not establish a causal mechanism.

Retrospective tail enrichment among exchanged candidates. For each V2 task-split-policy-seed-stage, we contrasted candidates selected only by LOCAL with candidates selected only by RAW; the intersection selected by both is counted separately. Stage 0 uses an identical candidate pool, while later stages compare each policy’s actual batch after its own preceding trajectory. Target-pool tail membership is taken from the stored, post-selection labels and was

not used during acquisition. Table S8 reports pooled entrant counts and tail fractions over 20 seeds and four stages. The rates are descriptive candidate-level proportions; the eight task-split environments are summarized with equal weight in the pooled overview, and no bootstrap intervals or significance tests are used. Mean target value is reported only as a secondary descriptive quantity in the machine-readable source table; it is not a substitute for target-tail enrichment.

10.2 Secondary sensitivity to the interpolation exponent α

The frozen V2 setting remains $\alpha = 0.5$. To assess whether the qualitative task-policy pattern depends on this value, a separate secondary analysis ran fresh sequential LOCAL campaigns at $\alpha \in \{0.25, 0.50, 0.75, 1.00\}$ in four prespecified representative environments, under both acquisition policies and all 20 V2 seeds (640 LOCAL trajectories). The 160 RAW references are the hash-verified V2 campaigns because their scores and decisions do not depend on α . The alpha=0.50 trajectories were rerun rather than rescored from stored states, and were checked against the frozen V2 LOCAL trajectory. No alpha was selected or tuned. Table S9 shows environment-level top-tail summaries and Figure S7 shows the mean top-1% effect by environment. Seed ranges are observed minima and maxima, not confidence intervals; no hypothesis tests were performed. This analysis evaluates robustness only and does not change the primary alpha=0.5 result.

Table S9: Alpha sensitivity across the four prespecified V2 environments.

α	Task	Split	Policy	Tail	Mean Δ	SD	Seed range	Positive/ n
0.25	Expt gap	IID	Magnitude	0.5%	-0.015	0.041	[-0.127, +0.065]	7/20
0.25	Expt gap	IID	Magnitude	1%	-0.026	0.043	[-0.138, +0.068]	4/20
0.25	Expt gap	IID	Magnitude	5%	-0.002	0.036	[-0.132, +0.060]	12/20
0.25	Expt gap	IID	Rank	0.5%	-0.034	0.073	[-0.197, +0.047]	7/20
0.25	Expt gap	IID	Rank	1%	-0.020	0.048	[-0.109, +0.102]	5/20
0.25	Expt gap	IID	Rank	5%	+0.007	0.029	[-0.029, +0.079]	11/20
0.25	Log bulk (KVRH)	chem OOD	Magnitude	0.5%	+0.035	0.070	[-0.124, +0.177]	16/20
0.25	Log bulk (KVRH)	chem OOD	Magnitude	1%	+0.035	0.065	[-0.128, +0.172]	16/20
0.25	Log bulk (KVRH)	chem OOD	Magnitude	5%	-0.016	0.055	[-0.140, +0.082]	9/20
0.25	Log bulk (KVRH)	chem OOD	Rank	0.5%	+0.056	0.108	[-0.090, +0.287]	11/20
0.25	Log bulk (KVRH)	chem OOD	Rank	1%	+0.053	0.086	[-0.044, +0.263]	14/20
0.25	Log bulk (KVRH)	chem OOD	Rank	5%	+0.021	0.040	[-0.038, +0.112]	14/20
0.25	MP gap	chem OOD	Magnitude	0.5%	-0.004	0.029	[-0.078, +0.045]	10/20
0.25	MP gap	chem OOD	Magnitude	1%	-0.000	0.040	[-0.087, +0.056]	11/20
0.25	MP gap	chem OOD	Magnitude	5%	+0.005	0.032	[-0.048, +0.058]	12/20
0.25	MP gap	chem OOD	Rank	0.5%	-0.039	0.036	[-0.102, +0.017]	2/20
0.25	MP gap	chem OOD	Rank	1%	-0.027	0.034	[-0.086, +0.030]	4/20
0.25	MP gap	chem OOD	Rank	5%	+0.012	0.029	[-0.047, +0.054]	13/20
0.25	MP gap	IID	Magnitude	0.5%	-0.004	0.015	[-0.033, +0.027]	7/20
0.25	MP gap	IID	Magnitude	1%	+0.002	0.011	[-0.026, +0.019]	12/20
0.25	MP gap	IID	Magnitude	5%	+0.010	0.009	[-0.008, +0.028]	19/20
0.25	MP gap	IID	Rank	0.5%	-0.031	0.021	[-0.071, +0.017]	1/20
0.25	MP gap	IID	Rank	1%	-0.018	0.024	[-0.060, +0.021]	5/20
0.25	MP gap	IID	Rank	5%	+0.015	0.014	[-0.006, +0.040]	17/20
0.50	Expt gap	IID	Magnitude	0.5%	+0.024	0.081	[-0.134, +0.183]	11/20
0.50	Expt gap	IID	Magnitude	1%	+0.020	0.058	[-0.074, +0.169]	14/20
0.50	Expt gap	IID	Magnitude	5%	+0.017	0.040	[-0.080, +0.110]	15/20
0.50	Expt gap	IID	Rank	0.5%	-0.035	0.064	[-0.188, +0.092]	6/20
0.50	Expt gap	IID	Rank	1%	-0.024	0.040	[-0.094, +0.044]	5/20
0.50	Expt gap	IID	Rank	5%	+0.009	0.032	[-0.048, +0.099]	15/20
0.50	Log bulk (KVRH)	chem OOD	Magnitude	0.5%	+0.050	0.074	[-0.155, +0.185]	17/20
0.50	Log bulk (KVRH)	chem OOD	Magnitude	1%	+0.042	0.080	[-0.160, +0.172]	17/20
0.50	Log bulk (KVRH)	chem OOD	Magnitude	5%	-0.044	0.070	[-0.176, +0.068]	6/20

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Table S9: Alpha sensitivity across the four prespecified V2 environments (continued).

α	Task	Split	Policy	Tail	Mean Δ	SD	Seed range	Positive/ n
0.50	Log bulk (KVRH)	chem OOD	Rank	0.5%	+0.068	0.155	[-0.278, +0.333]	14/20
0.50	Log bulk (KVRH)	chem OOD	Rank	1%	+0.057	0.119	[-0.268, +0.248]	15/20
0.50	Log bulk (KVRH)	chem OOD	Rank	5%	+0.034	0.059	[-0.137, +0.135]	16/20
0.50	MP gap	chem OOD	Magnitude	0.5%	-0.019	0.039	[-0.086, +0.044]	6/20
0.50	MP gap	chem OOD	Magnitude	1%	-0.008	0.049	[-0.091, +0.061]	8/20
0.50	MP gap	chem OOD	Magnitude	5%	+0.002	0.039	[-0.064, +0.080]	10/20
0.50	MP gap	chem OOD	Rank	0.5%	-0.071	0.048	[-0.162, +0.026]	2/20
0.50	MP gap	chem OOD	Rank	1%	-0.040	0.052	[-0.141, +0.086]	2/20
0.50	MP gap	chem OOD	Rank	5%	+0.018	0.038	[-0.041, +0.096]	16/20
0.50	MP gap	IID	Magnitude	0.5%	-0.016	0.027	[-0.057, +0.029]	7/20
0.50	MP gap	IID	Magnitude	1%	-0.005	0.018	[-0.039, +0.019]	7/20
0.50	MP gap	IID	Magnitude	5%	+0.007	0.012	[-0.026, +0.020]	15/20
0.50	MP gap	IID	Rank	0.5%	-0.067	0.037	[-0.155, +0.011]	1/20
0.50	MP gap	IID	Rank	1%	-0.041	0.035	[-0.132, +0.020]	3/20
0.50	MP gap	IID	Rank	5%	+0.015	0.017	[-0.020, +0.038]	16/20
0.75	Expt gap	IID	Magnitude	0.5%	+0.038	0.096	[-0.146, +0.256]	13/20
0.75	Expt gap	IID	Magnitude	1%	+0.025	0.065	[-0.086, +0.154]	12/20
0.75	Expt gap	IID	Magnitude	5%	+0.015	0.054	[-0.061, +0.175]	12/20
0.75	Expt gap	IID	Rank	0.5%	-0.039	0.093	[-0.243, +0.150]	6/20
0.75	Expt gap	IID	Rank	1%	-0.033	0.064	[-0.129, +0.075]	7/20
0.75	Expt gap	IID	Rank	5%	+0.007	0.034	[-0.065, +0.081]	13/20
0.75	Log bulk (KVRH)	chem OOD	Magnitude	0.5%	+0.006	0.127	[-0.339, +0.232]	12/20
0.75	Log bulk (KVRH)	chem OOD	Magnitude	1%	-0.023	0.119	[-0.357, +0.179]	9/20
0.75	Log bulk (KVRH)	chem OOD	Magnitude	5%	-0.095	0.090	[-0.291, +0.033]	4/20
0.75	Log bulk (KVRH)	chem OOD	Rank	0.5%	+0.052	0.143	[-0.176, +0.381]	11/20
0.75	Log bulk (KVRH)	chem OOD	Rank	1%	+0.058	0.106	[-0.181, +0.238]	14/20
0.75	Log bulk (KVRH)	chem OOD	Rank	5%	+0.035	0.060	[-0.121, +0.134]	14/20
0.75	MP gap	chem OOD	Magnitude	0.5%	-0.037	0.047	[-0.119, +0.048]	4/20
0.75	MP gap	chem OOD	Magnitude	1%	-0.028	0.047	[-0.106, +0.058]	5/20
0.75	MP gap	chem OOD	Magnitude	5%	-0.018	0.045	[-0.088, +0.064]	6/20
0.75	MP gap	chem OOD	Rank	0.5%	-0.090	0.056	[-0.241, +0.003]	1/20
0.75	MP gap	chem OOD	Rank	1%	-0.050	0.062	[-0.200, +0.081]	3/20
0.75	MP gap	chem OOD	Rank	5%	+0.019	0.039	[-0.058, +0.089]	15/20
0.75	MP gap	IID	Magnitude	0.5%	-0.038	0.043	[-0.116, +0.031]	5/20
0.75	MP gap	IID	Magnitude	1%	-0.022	0.031	[-0.088, +0.022]	6/20
0.75	MP gap	IID	Magnitude	5%	-0.007	0.016	[-0.049, +0.018]	7/20
0.75	MP gap	IID	Rank	0.5%	-0.077	0.034	[-0.153, -0.000]	0/20
0.75	MP gap	IID	Rank	1%	-0.047	0.033	[-0.111, +0.009]	2/20
0.75	MP gap	IID	Rank	5%	+0.023	0.022	[-0.015, +0.060]	17/20
1.00	Expt gap	IID	Magnitude	0.5%	+0.008	0.108	[-0.171, +0.223]	12/20
1.00	Expt gap	IID	Magnitude	1%	+0.002	0.075	[-0.111, +0.214]	10/20
1.00	Expt gap	IID	Magnitude	5%	-0.003	0.066	[-0.096, +0.165]	8/20
1.00	Expt gap	IID	Rank	0.5%	-0.061	0.098	[-0.253, +0.079]	7/20
1.00	Expt gap	IID	Rank	1%	-0.044	0.076	[-0.182, +0.060]	8/20
1.00	Expt gap	IID	Rank	5%	+0.004	0.033	[-0.051, +0.078]	10/20
1.00	Log bulk (KVRH)	chem OOD	Magnitude	0.5%	-0.028	0.168	[-0.560, +0.220]	9/20
1.00	Log bulk (KVRH)	chem OOD	Magnitude	1%	-0.070	0.159	[-0.535, +0.169]	6/20
1.00	Log bulk (KVRH)	chem OOD	Magnitude	5%	-0.147	0.098	[-0.373, -0.006]	0/20
1.00	Log bulk (KVRH)	chem OOD	Rank	0.5%	+0.097	0.124	[-0.209, +0.301]	17/20
1.00	Log bulk (KVRH)	chem OOD	Rank	1%	+0.087	0.102	[-0.196, +0.267]	19/20
1.00	Log bulk (KVRH)	chem OOD	Rank	5%	+0.042	0.056	[-0.117, +0.151]	17/20
1.00	MP gap	chem OOD	Magnitude	0.5%	-0.083	0.060	[-0.182, -0.005]	0/20
1.00	MP gap	chem OOD	Magnitude	1%	-0.062	0.062	[-0.144, +0.071]	2/20
1.00	MP gap	chem OOD	Magnitude	5%	-0.053	0.048	[-0.132, +0.046]	3/20
1.00	MP gap	chem OOD	Rank	0.5%	-0.097	0.065	[-0.251, +0.024]	1/20
1.00	MP gap	chem OOD	Rank	1%	-0.053	0.064	[-0.214, +0.071]	3/20
1.00	MP gap	chem OOD	Rank	5%	+0.025	0.040	[-0.048, +0.110]	13/20
1.00	MP gap	IID	Magnitude	0.5%	-0.062	0.051	[-0.150, +0.035]	3/20
1.00	MP gap	IID	Magnitude	1%	-0.049	0.039	[-0.127, +0.008]	2/20
1.00	MP gap	IID	Magnitude	5%	-0.039	0.024	[-0.088, +0.001]	2/20
1.00	MP gap	IID	Rank	0.5%	-0.085	0.040	[-0.140, +0.001]	1/20
1.00	MP gap	IID	Rank	1%	-0.055	0.036	[-0.108, +0.007]	2/20

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Table S9: Alpha sensitivity across the four prespecified V2 environments (continued).

α	Task	Split	Policy	Tail	Mean Δ	SD	Seed range	Positive/ n
1.00	MP gap	IID	Rank	5%	+0.018	0.023	[-0.030, +0.064]	16/20

Across environments, top-1% rank differences stayed small and negative (-0.003 to -0.018); magnitude changed from $+0.003/+0.012$ at $\alpha = 0.25/0.50$ to $-0.012/-0.045$ at $0.75/1.00$. At 5%, rank stayed positive ($+0.014$ to $+0.022$), while magnitude reached -0.061 at $\alpha = 1.00$. No new exponent is selected; $\alpha = 0.50$ remains primary.

Table S10: Task-split V2 effects across target-tail definitions. The estimate is LOCAL minus RAW integrated recall $L_{4,\text{query}}$.

Task	Split	Policy	Tail	Mean $\Delta L_{4,\text{query}}$	SD	Seed range	Positive
Expt. gap	IID	Rank	0.5%	-0.035	0.064	[-0.188, 0.092]	6/20
Expt. gap	IID	Rank	1%	-0.024	0.040	[-0.094, 0.044]	5/20
Expt. gap	IID	Rank	5%	+0.009	0.032	[-0.048, 0.099]	15/20
Expt. gap	IID	Magnitude	0.5%	+0.024	0.081	[-0.134, 0.183]	11/20
Expt. gap	IID	Magnitude	1%	+0.020	0.058	[-0.074, 0.169]	14/20
Expt. gap	IID	Magnitude	5%	+0.017	0.040	[-0.080, 0.110]	15/20
Expt. gap	Chem. OOD	Rank	0.5%	-0.002	0.102	[-0.212, 0.248]	12/20
Expt. gap	Chem. OOD	Rank	1%	-0.005	0.103	[-0.264, 0.193]	11/20
Expt. gap	Chem. OOD	Rank	5%	+0.015	0.080	[-0.158, 0.154]	14/20
Expt. gap	Chem. OOD	Magnitude	0.5%	+0.027	0.133	[-0.222, 0.392]	13/20
Expt. gap	Chem. OOD	Magnitude	1%	+0.014	0.119	[-0.211, 0.282]	13/20
Expt. gap	Chem. OOD	Magnitude	5%	+0.019	0.082	[-0.223, 0.145]	14/20
MP gap	IID	Rank	0.5%	-0.067	0.037	[-0.155, 0.011]	1/20
MP gap	IID	Rank	1%	-0.041	0.035	[-0.132, 0.020]	3/20
MP gap	IID	Rank	5%	+0.015	0.017	[-0.020, 0.038]	16/20
MP gap	IID	Magnitude	0.5%	-0.016	0.027	[-0.057, 0.029]	7/20
MP gap	IID	Magnitude	1%	-0.005	0.018	[-0.039, 0.019]	7/20
MP gap	IID	Magnitude	5%	+0.007	0.012	[-0.026, 0.020]	15/20
MP gap	Chem. OOD	Rank	0.5%	-0.071	0.048	[-0.162, 0.026]	2/20
MP gap	Chem. OOD	Rank	1%	-0.040	0.052	[-0.141, 0.086]	2/20
MP gap	Chem. OOD	Rank	5%	+0.018	0.038	[-0.041, 0.096]	16/20
MP gap	Chem. OOD	Magnitude	0.5%	-0.019	0.039	[-0.086, 0.044]	6/20
MP gap	Chem. OOD	Magnitude	1%	-0.008	0.049	[-0.091, 0.061]	8/20
MP gap	Chem. OOD	Magnitude	5%	+0.002	0.039	[-0.064, 0.080]	10/20
log shear modulus (GVRH)	IID	Rank	0.5%	+0.013	0.107	[-0.224, 0.189]	12/20
log shear modulus (GVRH)	IID	Rank	1%	+0.009	0.084	[-0.198, 0.123]	11/20
log shear modulus (GVRH)	IID	Rank	5%	+0.003	0.032	[-0.079, 0.080]	8/20
log shear modulus (GVRH)	IID	Magnitude	0.5%	+0.028	0.055	[-0.082, 0.113]	15/20
log shear modulus (GVRH)	IID	Magnitude	1%	+0.022	0.069	[-0.107, 0.137]	14/20
log shear modulus (GVRH)	IID	Magnitude	5%	-0.010	0.045	[-0.091, 0.084]	10/20
log shear modulus (GVRH)	Chem. OOD	Rank	0.5%	+0.048	0.115	[-0.110, 0.252]	10/20
log shear modulus (GVRH)	Chem. OOD	Rank	1%	+0.054	0.106	[-0.068, 0.266]	11/20
log shear modulus (GVRH)	Chem. OOD	Rank	5%	+0.020	0.051	[-0.074, 0.127]	13/20
log shear modulus (GVRH)	Chem. OOD	Magnitude	0.5%	+0.030	0.071	[-0.115, 0.169]	14/20
log shear modulus (GVRH)	Chem. OOD	Magnitude	1%	+0.020	0.057	[-0.053, 0.158]	13/20
log shear modulus (GVRH)	Chem. OOD	Magnitude	5%	-0.008	0.039	[-0.081, 0.066]	9/20
log bulk modulus (KVRH)	IID	Rank	0.5%	+0.043	0.062	[-0.150, 0.127]	16/20
log bulk modulus (KVRH)	IID	Rank	1%	+0.037	0.049	[-0.098, 0.096]	16/20
log bulk modulus (KVRH)	IID	Rank	5%	+0.012	0.031	[-0.034, 0.061]	11/20
log bulk modulus (KVRH)	IID	Magnitude	0.5%	+0.023	0.075	[-0.102, 0.169]	12/20
log bulk modulus (KVRH)	IID	Magnitude	1%	+0.014	0.068	[-0.103, 0.139]	11/20
log bulk modulus (KVRH)	IID	Magnitude	5%	-0.071	0.054	[-0.158, 0.004]	1/20
log bulk modulus (KVRH)	Chem. OOD	Rank	0.5%	+0.068	0.155	[-0.278, 0.333]	14/20

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Table S10: Task-split V2 effects across target-tail definitions. The estimate is LOCAL minus RAW integrated recall $L_{4,\text{query}}$. (continued)

Task	Split	Policy	Tail	Mean $\Delta L_{4,\text{query}}$	SD	Seed range	Positive
log bulk modulus (KVRH)	Chem.	OOD Rank	1%	+0.057	0.119	[-0.268, 0.248]	15/20
log bulk modulus (KVRH)	Chem.	OOD Rank	5%	+0.034	0.059	[-0.137, 0.135]	16/20
log bulk modulus (KVRH)	Chem.	OOD Magnitude	0.5%	+0.050	0.074	[-0.155, 0.185]	17/20
log bulk modulus (KVRH)	Chem.	OOD Magnitude	1%	+0.042	0.080	[-0.160, 0.172]	17/20
log bulk modulus (KVRH)	Chem.	OOD Magnitude	5%	-0.044	0.070	[-0.176, 0.068]	6/20

Each row summarizes 20 paired seed blocks. The seed range is the observed minimum–maximum, not a confidence interval. Task-split means are descriptive and no V2 hypothesis tests were performed.

Table S11: V2 initial-batch selection disagreement between LOCAL and RAW.

Task	Split	Policy	Mean (%)	Median (%)	Range (%)
Expt. gap	IID	Rank	18.9	14.1	[2.2, 49.3]
Expt. gap	IID	Magnitude	29.3	23.6	[2.9, 66.7]
Expt. gap	Chem. OOD	Rank	21.7	19.2	[0.7, 73.9]
Expt. gap	Chem. OOD	Magnitude	24.4	23.6	[3.6, 58.7]
MP gap	IID	Rank	26.8	27.1	[19.2, 33.4]
MP gap	IID	Magnitude	24.7	25.0	[17.8, 36.9]
MP gap	Chem. OOD	Rank	32.0	31.7	[22.0, 43.7]
MP gap	Chem. OOD	Magnitude	26.8	26.6	[16.2, 41.8]
log shear modulus (GVRH)	IID	Rank	21.8	19.5	[9.4, 38.0]
log shear modulus (GVRH)	IID	Magnitude	32.2	31.9	[13.4, 56.5]
log shear modulus (GVRH)	Chem. OOD	Rank	25.5	25.5	[12.5, 40.7]
log shear modulus (GVRH)	Chem. OOD	Magnitude	37.9	35.6	[7.0, 64.4]
log bulk modulus (KVRH)	IID	Rank	28.5	24.9	[16.7, 41.9]
log bulk modulus (KVRH)	IID	Magnitude	48.2	44.5	[34.3, 80.9]
log bulk modulus (KVRH)	Chem. OOD	Rank	31.2	31.0	[14.0, 46.5]
log bulk modulus (KVRH)	Chem. OOD	Magnitude	53.0	55.5	[26.7, 84.5]

Mean and median summarize 20 fixed seed pairs per row. Disagreement is the fraction of RAW first-batch candidates absent from the LOCAL first batch. Values are descriptive; no independent-task inference is implied.

Table S12: V2 diagnostic-partition MACE by task, split, acquisition policy, and campaign condition.

Task	Split	Policy	Condition	RAW MACE	LOCAL MACE	LOCAL – RAW
Expt. gap	IID	Rank	RAW	0.055	0.062	+0.008
Expt. gap	IID	Rank	LOCAL	0.056	0.065	+0.009
Expt. gap	IID	Magnitude	RAW	0.055	0.064	+0.009
Expt. gap	IID	Magnitude	LOCAL	0.054	0.062	+0.008
Expt. gap	Chem. OOD	Rank	RAW	0.054	0.073	+0.018
Expt. gap	Chem. OOD	Rank	LOCAL	0.053	0.069	+0.016
Expt. gap	Chem. OOD	Magnitude	RAW	0.055	0.073	+0.018
Expt. gap	Chem. OOD	Magnitude	LOCAL	0.056	0.075	+0.018
MP gap	IID	Rank	RAW	0.030	0.048	+0.018
MP gap	IID	Rank	LOCAL	0.031	0.047	+0.016
MP gap	IID	Magnitude	RAW	0.031	0.047	+0.016
MP gap	IID	Magnitude	LOCAL	0.031	0.046	+0.015
MP gap	Chem. OOD	Rank	RAW	0.033	0.054	+0.020
MP gap	Chem. OOD	Rank	LOCAL	0.035	0.051	+0.017
MP gap	Chem. OOD	Magnitude	RAW	0.034	0.049	+0.015

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Table S12: V2 diagnostic-partition MACE by task, split, acquisition policy, and campaign condition. (continued)

Task	Split	Policy	Condition	RAW MACE	LOCAL MACE	LOCAL – RAW
MP gap	Chem. OOD	Magnitude	LOCAL	0.035	0.048	+0.013
Log shear (GVRH)	IID	Rank	RAW	0.098	0.045	-0.053
Log shear (GVRH)	IID	Rank	LOCAL	0.094	0.044	-0.051
Log shear (GVRH)	IID	Magnitude	RAW	0.103	0.047	-0.056
Log shear (GVRH)	IID	Magnitude	LOCAL	0.107	0.040	-0.067
Log shear (GVRH)	Chem. OOD	Rank	RAW	0.077	0.051	-0.026
Log shear (GVRH)	Chem. OOD	Rank	LOCAL	0.069	0.053	-0.016
Log shear (GVRH)	Chem. OOD	Magnitude	RAW	0.086	0.048	-0.038
Log shear (GVRH)	Chem. OOD	Magnitude	LOCAL	0.075	0.051	-0.025
Log bulk (KVRH)	IID	Rank	RAW	0.095	0.042	-0.053
Log bulk (KVRH)	IID	Rank	LOCAL	0.083	0.043	-0.040
Log bulk (KVRH)	IID	Magnitude	RAW	0.101	0.043	-0.058
Log bulk (KVRH)	IID	Magnitude	LOCAL	0.095	0.044	-0.051
Log bulk (KVRH)	Chem. OOD	Rank	RAW	0.084	0.042	-0.043
Log bulk (KVRH)	Chem. OOD	Rank	LOCAL	0.074	0.043	-0.031
Log bulk (KVRH)	Chem. OOD	Magnitude	RAW	0.088	0.044	-0.044
Log bulk (KVRH)	Chem. OOD	Magnitude	LOCAL	0.084	0.042	-0.042

Each row averages 20 seed runs; each run-level value averages four campaign stages. MACE is the mean absolute difference between empirical and nominal coverage on the fixed diagnostic partition at 50%, 80%, 90%, and 95%. This is an auxiliary descriptive diagnostic, not a target-pool utility endpoint.

Table S13: Exploratory V2 candidate-level diagnostics for matbench_mp_gap.

Split	Policy	Condition	$\rho(\sigma_c, \epsilon)$	Mean rank shift	Top-5% σ ratio	Entrant tail (%)
IID	Rank	RAW	0.759	1948.2	2.050	18.1
IID	Rank	LOCAL	0.747	2068.8	1.944	16.8
IID	Magnitude	RAW	0.758	1613.3	2.347	14.6
IID	Magnitude	LOCAL	0.750	1611.2	2.176	13.7
Chem. OOD	Rank	RAW	0.699	2285.7	1.593	17.4
Chem. OOD	Rank	LOCAL	0.686	2525.5	1.538	15.4
Chem. OOD	Magnitude	RAW	0.693	2002.2	1.714	13.6
Chem. OOD	Magnitude	LOCAL	0.687	1979.3	1.617	13.5

Rows summarize 80 run stages (20 seeds $\times$ four stages) with equal stage weighting. $\rho(\sigma_c, |\epsilon|)$ is Spearman correlation between corrected uncertainty and absolute prediction error; mean rank shift compares counterfactual RAW and LOCAL candidate ranks. The Top-5% ratio is the ratio of mean stage medians for corrected uncertainty among target-pool top-5% candidates versus other candidates. Entrant tail is the fraction of LOCAL-only batch entrants in the top-5% target tail. These retrospective diagnostics are exploratory and do not establish a causal mechanism.

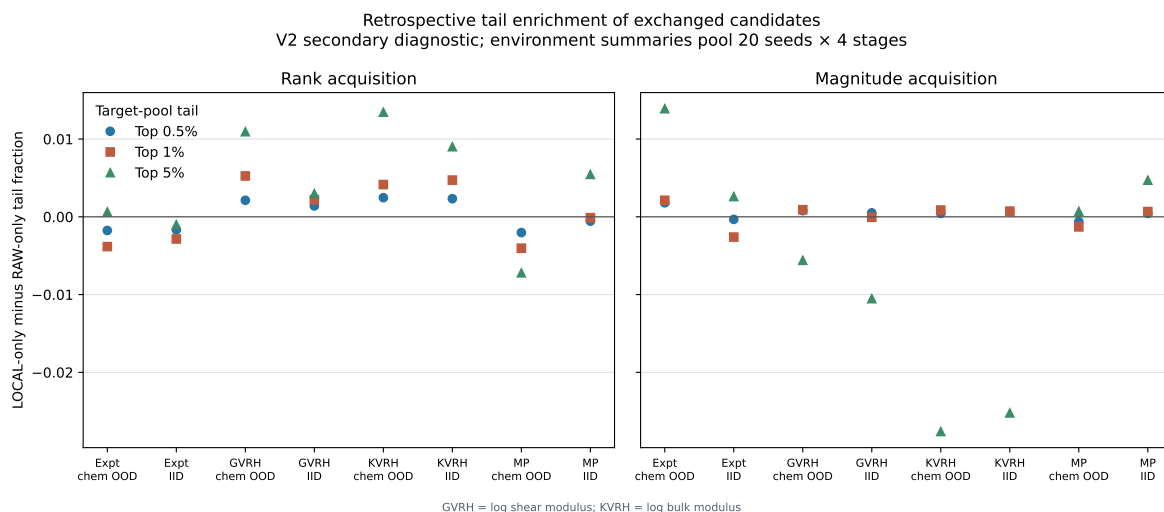


Figure S6: V2 retrospective difference between the target-tail fraction among LOCAL-only entrants and RAW-only entrants for each of the eight fixed task-split environments. Panels separate rank and magnitude acquisition; marker color and shape distinguish target-pool tails of 0.5%, 1%, and 5%. Each environment summary pools 20 paired seeds and four campaign stages. The horizontal line marks zero; positive values indicate greater tail enrichment among LOCAL-only entrants. Points have no interval bars and are descriptive secondary diagnostics.

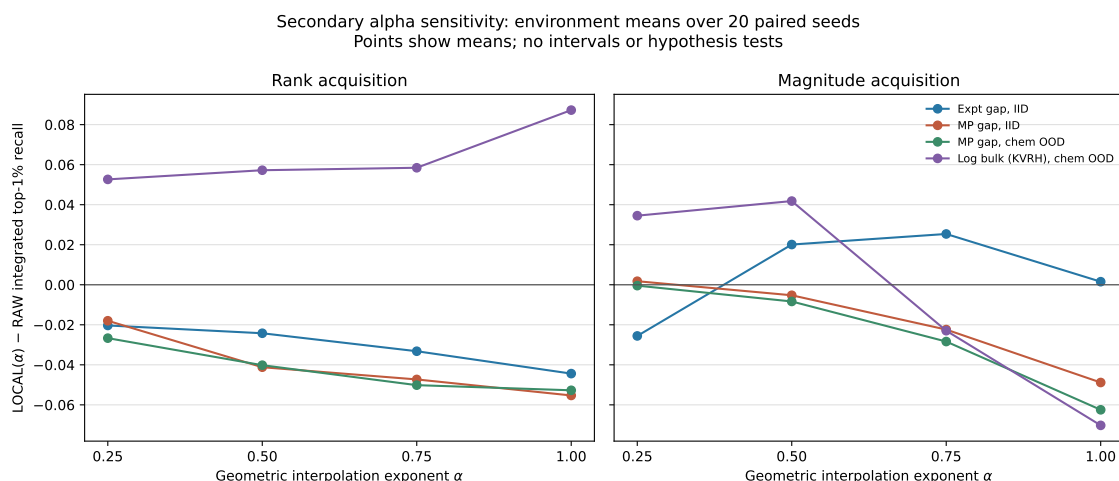


Figure S7: Secondary sequential alpha sensitivity. Lines show the mean LOCAL(α)-RAW integrated top-1% recall difference over 20 paired seeds in each of four prespecified task-split environments. Panels show rank and magnitude acquisition. The horizontal line marks zero; no intervals or hypothesis tests are shown. The $\alpha=0.50$ value is the frozen V2 setting, and this analysis does not select a replacement.

Table S8: V2 exchanged-candidate counts and retrospective target-tail enrichment.

Task	Split	Policy	Tail	LOCAL-only n	RAW-only n	Both n	LOCAL tail	RAW tail	Δ	Ratio
Expt gap	chemical OOD	Magnitude	0.5%	6116	6116	4924	0.014	0.012	+0.002	1.15
Expt gap	chemical OOD	Magnitude	1%	6116	6116	4924	0.026	0.024	+0.002	1.09
Expt gap	chemical OOD	Magnitude	5%	6116	6116	4924	0.129	0.115	+0.014	1.12
Expt gap	IID	Magnitude	0.5%	6132	6132	4908	0.011	0.011	-0.000	0.97
Expt gap	IID	Magnitude	1%	6132	6132	4908	0.022	0.025	-0.003	0.90
Expt gap	IID	Magnitude	5%	6132	6132	4908	0.128	0.126	+0.003	1.02
Log shear (GVRH)	chemical OOD	Magnitude	0.5%	16459	16459	9901	0.013	0.012	+0.001	1.07
Log shear (GVRH)	chemical OOD	Magnitude	1%	16459	16459	9901	0.027	0.026	+0.001	1.03
Log shear (GVRH)	chemical OOD	Magnitude	5%	16459	16459	9901	0.158	0.163	-0.006	0.97
Log shear (GVRH)	IID	Magnitude	0.5%	15676	15676	10704	0.014	0.013	+0.001	1.04
Log shear (GVRH)	IID	Magnitude	1%	15676	15676	10704	0.029	0.029	-0.000	1.00
Log shear (GVRH)	IID	Magnitude	5%	15676	15676	10704	0.150	0.161	-0.011	0.93
Log bulk (KVRH)	chemical OOD	Magnitude	0.5%	18438	18438	7922	0.018	0.018	+0.000	1.02
Log bulk (KVRH)	chemical OOD	Magnitude	1%	18438	18438	7922	0.038	0.037	+0.001	1.02
Log bulk (KVRH)	chemical OOD	Magnitude	5%	18438	18438	7922	0.164	0.192	-0.028	0.86
Log bulk (KVRH)	IID	Magnitude	0.5%	17886	17886	8494	0.016	0.015	+0.001	1.04
Log bulk (KVRH)	IID	Magnitude	1%	17886	17886	8494	0.032	0.031	+0.001	1.02
Log bulk (KVRH)	IID	Magnitude	5%	17886	17886	8494	0.164	0.189	-0.025	0.87
MP gap	chemical OOD	Magnitude	0.5%	138869	138869	115811	0.012	0.013	-0.001	0.95
MP gap	chemical OOD	Magnitude	1%	138869	138869	115811	0.028	0.029	-0.001	0.96
MP gap	chemical OOD	Magnitude	5%	138869	138869	115811	0.177	0.177	+0.001	1.00
MP gap	IID	Magnitude	0.5%	127149	127149	127511	0.012	0.012	+0.000	1.04
MP gap	IID	Magnitude	1%	127149	127149	127511	0.027	0.026	+0.001	1.03
MP gap	IID	Magnitude	5%	127149	127149	127511	0.169	0.164	+0.005	1.03
Expt gap	chemical OOD	Rank	0.5%	6266	6266	4774	0.010	0.012	-0.002	0.86
Expt gap	chemical OOD	Rank	1%	6266	6266	4774	0.022	0.026	-0.004	0.85
Expt gap	chemical OOD	Rank	5%	6266	6266	4774	0.124	0.124	+0.001	1.01
Expt gap	IID	Rank	0.5%	5966	5966	5074	0.008	0.010	-0.002	0.83
Expt gap	IID	Rank	1%	5966	5966	5074	0.017	0.020	-0.003	0.85
Expt gap	IID	Rank	5%	5966	5966	5074	0.123	0.124	-0.001	0.99
Log shear (GVRH)	chemical OOD	Rank	0.5%	15992	15992	10368	0.014	0.012	+0.002	1.17
Log shear (GVRH)	chemical OOD	Rank	1%	15992	15992	10368	0.029	0.024	+0.005	1.22
Log shear (GVRH)	chemical OOD	Rank	5%	15992	15992	10368	0.148	0.137	+0.011	1.08
Log shear (GVRH)	IID	Rank	0.5%	15037	15037	11343	0.011	0.010	+0.001	1.15
Log shear (GVRH)	IID	Rank	1%	15037	15037	11343	0.023	0.021	+0.002	1.11
Log shear (GVRH)	IID	Rank	5%	15037	15037	11343	0.122	0.119	+0.003	1.03
Log bulk (KVRH)	chemical OOD	Rank	0.5%	16644	16644	9716	0.016	0.014	+0.002	1.18
Log bulk (KVRH)	chemical OOD	Rank	1%	16644	16644	9716	0.031	0.026	+0.004	1.16
Log bulk (KVRH)	chemical OOD	Rank	5%	16644	16644	9716	0.137	0.124	+0.013	1.11
Log bulk (KVRH)	IID	Rank	0.5%	16306	16306	10074	0.013	0.011	+0.002	1.21
Log bulk (KVRH)	IID	Rank	1%	16306	16306	10074	0.028	0.023	+0.005	1.21
Log bulk (KVRH)	IID	Rank	5%	16306	16306	10074	0.125	0.116	+0.009	1.08
MP gap	chemical OOD	Rank	0.5%	161629	161629	93051	0.013	0.015	-0.002	0.86
MP gap	chemical OOD	Rank	1%	161629	161629	93051	0.026	0.031	-0.004	0.87
MP gap	chemical OOD	Rank	5%	161629	161629	93051	0.161	0.168	-0.007	0.96
MP gap	IID	Rank	0.5%	145201	145201	109459	0.015	0.015	-0.001	0.96
MP gap	IID	Rank	1%	145201	145201	109459	0.030	0.030	-0.000	1.00
MP gap	IID	Rank	5%	145201	145201	109459	0.165	0.160	+0.005	1.03

Counts pool all 20 seeds and four sequential stages within each fixed task–split–policy. “Both” is the number selected by both conditions at that stage. Tail fractions are the proportions of LOCAL-only or RAW-only candidates belonging to the indicated target-pool tail. Δ is LOCAL-only minus RAW-only tail fraction; the enrichment ratio is their quotient when the RAW-only rate is nonzero. A zero or undefined denominator is not assigned a finite ratio. These retrospective labels were not used for candidate selection.

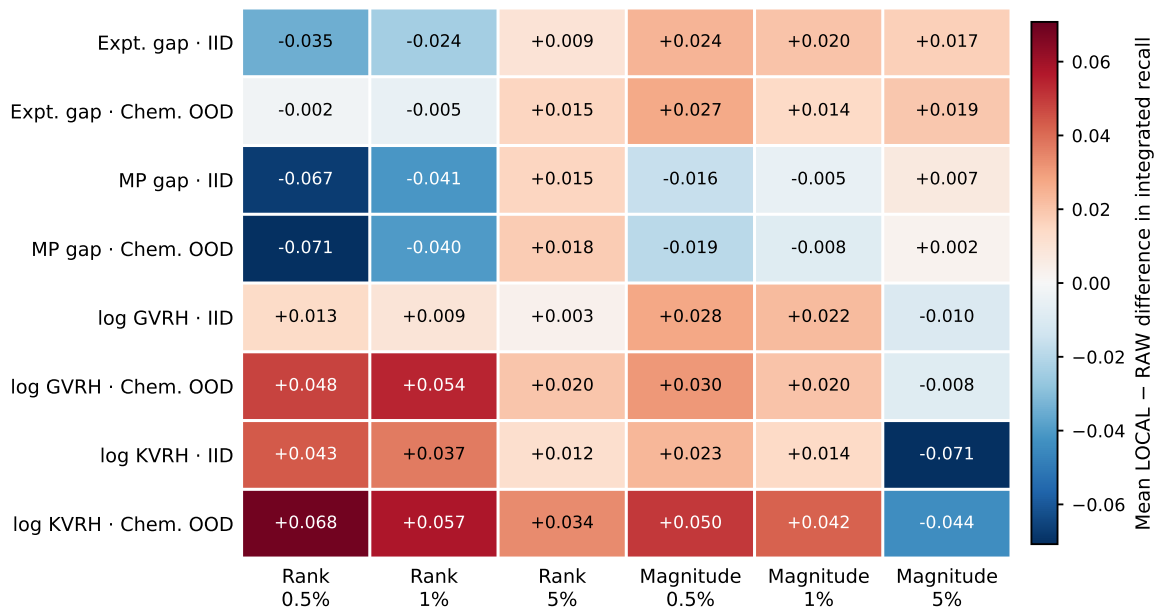


Figure S8: V2 mean LOCAL-RAW differences in integrated target-tail recall for each fixed task-split environment and policy-tail combination. The diverging color scale is centered at zero; annotated values are mean paired differences over 20 seeds. Positive values favor LOCAL. The heatmap is descriptive and does not show seed-level uncertainty or support task-level inferential claims.

Table S14: Exploratory V2 mean target values among queried candidates.

Task	Split	Policy	RAW	LOCAL	LOCAL – RAW	SD
Expt. gap	IID	Rank	1.885	1.952	+0.067	0.148
Expt. gap	IID	Magnitude	1.992	1.927	-0.065	0.171
Expt. gap	Chem. OOD	Rank	1.772	1.844	+0.072	0.175
Expt. gap	Chem. OOD	Magnitude	1.855	1.879	+0.024	0.172
MP gap	IID	Rank	2.908	2.929	+0.021	0.050
MP gap	IID	Magnitude	3.205	3.167	-0.038	0.031
MP gap	Chem. OOD	Rank	2.694	2.698	+0.004	0.090
MP gap	Chem. OOD	Magnitude	2.954	2.939	-0.015	0.076
log shear modulus (GVRH)	IID	Rank	1.786	1.792	+0.006	0.021
log shear modulus (GVRH)	IID	Magnitude	1.872	1.812	-0.060	0.022
log shear modulus (GVRH)	Chem. OOD	Rank	1.797	1.796	0.000	0.035
log shear modulus (GVRH)	Chem. OOD	Magnitude	1.871	1.809	-0.063	0.044
log bulk modulus (KVRH)	IID	Rank	2.084	2.094	+0.010	0.034
log bulk modulus (KVRH)	IID	Magnitude	2.185	2.080	-0.105	0.038
log bulk modulus (KVRH)	Chem. OOD	Rank	2.066	2.070	+0.003	0.025
log bulk modulus (KVRH)	Chem. OOD	Magnitude	2.171	2.068	-0.103	0.039

Each row summarizes 20 paired seed blocks. This retrospective descriptive average is not top-tail enrichment and is not used as a confirmatory endpoint.

References

- Leo Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001. doi: 10.1023/A:1010933404324.
- Shyue Ping Ong, William Davidson Richards, Anubhav Jain, Geoffroy Hautier, Michael Kocher, Shreyas Cholia, Dan Gunter, Vincent L. Chevrier, Kristin A. Persson, and Gerbrand Ceder. Python materials genomics (pymatgen): A robust, open-source python library for materials analysis. *Computational Materials Science*, 68:314–319, 2013. doi: 10.1016/j.commatsci.2012.10.028.